Tesseract: Self-Replicating Code Organisms That Evolve Their Own Attention Layers

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Abstract

Tesseract is an artificial-life system in which every organism is a file that carries its own genome. In the expression world the genome is an arithmetic program; in the neural world it is a small transformer with its weights, architecture and a replay memory. Organisms earn energy on a habitat drawn from the Mandelbrot set, reproduce with mutation, and may prey on foreign neighbours, taking their energy and their code or brain. The neural task is induction, which a one-layer transformer cannot learn (0.14 held-out accuracy against 1.00 for two layers). We report three results. First, a function-preserving morph lets the architecture mutate without erasing what was learned, and three-layer lineages then appear and persist. Second, a kin rule that looked 24 ancestors deep made predation impossible, because the whole population descends from one root; limiting kinship to parents and grandparents restores hunting. Third, against two controls (random survival, and a single brain trained alone) selection adds nothing measurable on this task. Code, runs and the original version are public.

1 Introduction

Tierra and Avida showed that self-replicating programs competing for resources evolve parasites, cooperation and complex functions [1, 2, 3]. Neuroevolution asks a related question of networks: NEAT grows topologies from minimal starts [4], large-scale and regularized evolution search architectures while children inherit their parents' weights [5, 6], and population-based training copies trained weights between workers while it mutates hyperparameters [7]. Learning within a lifetime can also steer evolution [8].

Tesseract joins these lines in a deliberately small setting. An organism is a file. It reads its own genome, is scored by the environment, and writes mutated children; in the neural world it carries a transformer [9] that it trains during its life and passes on. We chose induction because depth is causal there: copying the token that followed an earlier match needs one head that tags each position with its predecessor and a second that reads the tag [10, 11]. A population that loses its second layer should fail.

The contribution is less a success story than an audit: building the controls changed what we can claim. Section 2 describes the system, Section 3 three findings, and Section 4 what they imply for evolving learners.

2 System

Organisms. In the expression world each organism is a Python file whose genome is one base64 JSON line: an expression of at most 240 characters, a habitat coordinate, copy fidelity μ, metabolic efficiency e, brood size and diet (the probability of hunting). The file reads itself, evaluates its expression on six integer lists (target: the sum of squares of the even entries) and, to reproduce, rewrites its own genome line with character-level mutations and writes the result to disk as a child. In the neural world the genome also holds an architecture (width d, MLP multiplier, layers L, learning rate), float32 weights and up to 32 replayed examples. Each generation an organism trains for 800 steps of clipped SGD on fresh batches plus its memory and is scored on 128 sequences it has never seen. Backpropagation is hand-written in NumPy and matches finite differences to 4×10^{-10}. A standard-library-only organism ($d=8$, 52 times slower) trains and rewrites itself on any machine with Python.

Habitat and energy. Fertility at (c_r,c_i) is the Mandelbrot escape time scaled to $[0.35,2.75]$, or 0.30 inside the set, so only the boundary is rich [12]; the iteration cap grows by three per generation. Energy changes by $pfe^{1/2}-6e^2$ for score p and fertility f, which puts the best efficiency at $e^*=(pf/24)^{2/3}$ with no bound on e. The $K/\bar{b}$ organisms with the most energy survive ($\bar{b}$ is the mean brood) and split their energy among their children.

Task. Eight distinct tokens from a vocabulary of 12 are followed by a repeat of one of them; the target is the token that came after it. After 800 steps a two-layer model ($d=32$) reaches 1.00 held-out accuracy and a one-layer model 0.14, the score of copying a random earlier token (1/7).

Morph. Architecture mutates in 2% of births ($L\pm1$, MLP multiplier ±1, or a new d). Instead of reinitialising, `morph()` embeds the old weights. An inserted block has $W_o=W_2=b_2=0$ and is therefore the identity on the residual stream, as with zero-initialised residual branches [13, 14]; new width reads from the stream but writes zeros into it, and W_q is scaled by $\sqrt{d'/d}$ to keep the attention temperature. Growth leaves every logit unchanged ($|\Delta|<10^{-15}$); shrinking discards weights.

Predation. With probability set by its diet gene an organism hunts: it pays 4 energy, takes 45% of a neighbour's energy, adopts the neighbour's expression or whole brain, and forages at 60% for that generation. Prey must not be kin, defined as sharing a parent or grandparent [15], and must be foreign: a different expression, or a relative weight distance of at least 0.10 (measured: siblings 0.06, second cousins 0.12, unrelated brains 1.4).

Containment. Expressions must pass an AST whitelist (arithmetic, comparisons, comprehensions, calls to nine built-ins; no attributes, imports or assignment) and run in a subprocess with a 6 s timeout and a 512 MB address-space limit, so `c*10**9` dies of `MemoryError` and its owner starves. Organisms have no network access.

3 Results

3.1 The predator that never ate

The original rule required prey to share no ancestor among the last 24 lineage entries. Every organism descends from one root whose identifier stays in every lineage for 24 generations, so no legal prey can exist before then, whatever the genotypes look like; afterwards a population of $N\approx12$ coalesces within $O(N)$ generations [16], so most pairs remain kin. Re-running the original code gave zero hunts in 34 generations of the expression world and in 12 of the neural world, and the earlier refusals had been read as a sign of clonality. With kinship

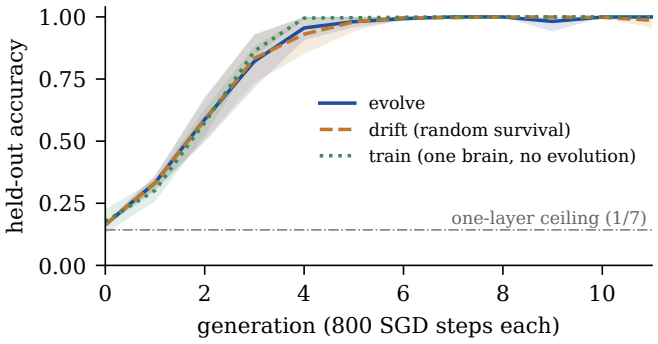


Figure 1: Mean held-out accuracy of the population per generation, seeds 7–11 (line: mean; band: ± 1 s.d.); $K = 12$, 800 SGD steps per generation.

limited to two generations the neural world made 5 to 19 hunts per 12-generation run (mean 12.6, five seeds), and an 80-generation expression run made 62 hunts while the diet gene rose from 0.05 to 0.40.

3.2 A layer born without forgetting

Before the morph, any change of shape reinitialised the brain, so a mutant raced parents with thousands of training steps and starved; a 2→1 layer mutation also crashed the simulation, because leftover weights passed the shape check. "Layers never dropped below two" therefore measured a handicap, not a necessity. With the morph, three-layer brains appeared in two of five runs: in one the line was born in generation 6 and three of the twelve newborns at the end carry it, in the other it died out. Under selection the single one-layer brain that was born died; under random survival one-layer brains appeared in three of five runs and one lasted to the last generation. On a task that two layers already solve, a third layer looks close to neutral, and losing the second is what selection removes.

3.3 Selection is not yet doing the work

Figure 1 compares the full world (*evolve*) with the same world under random survival (*drift*) and with a single brain trained alone, without mutation or selection (*train*). All three cross 0.95 mean held-out accuracy in generations 4 to 6 (mean over seeds: evolve 4.4, drift 4.6, train 4.0), after 3,200 to 4,800 SGD steps; at generation 3 they stand at 0.82 ± 0.11, 0.83 ± 0.10 and 0.86 ± 0.02. A single brain trained alone is as fast as the population. Inherited trained weights carry the learning; on this task selection rides along.

4 Discussion

Tesseract's lineages are Lamarckian: a child starts where its parent's training ended, as in population-based training [7]. When gradient descent can solve the task inside one lineage within a few generations, selection has nothing left to choose. Evolution should matter when a single learner cannot get there alone: tasks that shift, competing niches, or a compute budget too small for one lineage. Two lessons travel beyond this system. A kin-recognition window longer than the population's coalescence time makes everyone kin. And an architecture search that reinitialises its mutants measures the cost of forgetting, not the value of the architecture.

Limitations. Populations are small ($K = 12$); the neural results come from five seeds and one task, the expression-world numbers from single runs; retention is measured on examples the organism trained on.

Code and data. `https://github.com/myaslioglu/tesseract-alife`, archived at Zenodo (doi:10.5281/zenodo.XXXXXXX). Tag `v0` keeps the original code, and every number above can be regenerated with the commands in the README.

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